

Trainings ▾

General Information

The lubricating oil transfer tube connects the lubricating oil pump outlet to the oil transfer connection that mounts to the cylinder block. The oil transfer tube is different for K38 and K50 engines. The oil transfer tube is different depending on the location of the oil filter head. If the oil filter head is located on the right bank of the engine, the oil transfer tube has the shape of a "U". If the oil filter head is located on the left bank of the engine, the oil transfer tube is relatively straight. The oil transfer tube always causes the oil to flow from the oil pump to the engine bank opposite the engine bank where the lubricating oil filter head is mounted.

The oil transfer tube for the K50 low speed pump is **not** the same as the tube for the K50 high speed pump.

Older K50 oil transfer tubes were attached to the pump with 9.525 mm [0.375 inch] capscrews. The newer K50 oil transfer tubes are attached to the pump with 11.125 mm [0.438 inch] capscrews. The older tubes are **not** available for service. If the older tube **must** be replaced, the 9.525 mm [0.375 inch] capscREW to the pump **must** have a 9.525 mm [0.375 inch] and a 11.125 mm [0.438 inch] flat washer to be sure the smaller washer does **not** bend, allowing the capscREW to loosen.

The thickness of the transfer tubes was increased. It is recommended that the new thicker tube be installed during engine rebuild.

Lubricating Oil Transfer Tube to the Cylinder Block Connection:

The oil transfer tube to the cylinder block connection is different for K38 and K50 engines. The connection for the K38 engine does **not** contain a high pressure relief valve. The connection for the K50 engine has a high pressure relief valve when both the high speed and low speed pumps are installed.

Older K38 connections had a threaded hole for the attachment of a retainer shaped like a crowfoot for the transfer tube. Newer connections provide two threaded holes for attaching a sliding flange on the transfer tube. It is recommended that the new connection, Part Number 3026700, be installed during engine rebuild. Newer K38 cylinder blocks were modified for a connection, Part Number 3176713, that has the same capscREW hole spacing as the K50 engine. If the cylinder block is changed to the newer block, the connection **must** be replaced.

Lubricating Oil Transfer Tube Seals:

K38 engine have an o-ring seal on the ends of the oil transfer tubes. Some engines are manufactured with rectangular ring seals on the transfer tube. The rectangular seals on these engines **must** be replaced with o-ring seals when the transfer tube is removed.

K50 engines have a rectangular seal ring to seal the ends of the oil transfer tube. The seals on the transfer tube are **not** the same size as the oil suction tube.

The material for the manufacture of the o-ring seals for the K38 transfer tube and the rectangular seals for the K50 high speed pump transfer tube was changed. The new material has a higher temperature resistance. The new seals can be identified by a yellow stripe that is 13 mm [0.500 inch] wide.

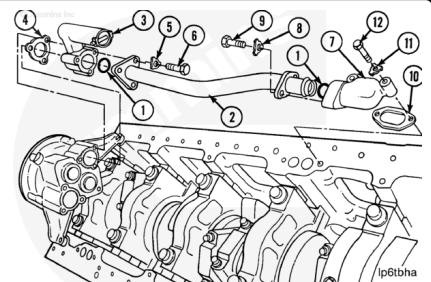
Remove

The oil transfer tubes for the K38 and K50 engines are also different. The tubes for the K50 low speed and high speed pumps are different.

The tubes are also different when the oil filter head is located on the right bank or left bank of the engine. **Only** the tubes for the left bank mounted filter head are shown. The procedure is the same if the tubes are for the right bank mounted filter head.

K38

- Bend all of the lockplates from the heads of the capscrews. Remove the capscrews (9) and the two capscrews (12). Remove the connection (7), the gasket (10), and the rectangular ring seal (1).
- Discard the seal and gasket.
- Remove the three capscrews (6). Remove the tube (2), the oil pressure relief valve (3), the gasket (4), and the seal (1).
- Discard the seal and gasket.



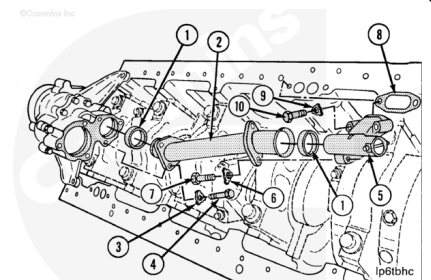
The Remove instructions for the K50 with Low Speed (Two Gear) Pump are listed below:

Bend all of the lockplates from the heads of the capscrews. Remove the two capscrews (7) and the two capscrews (10). Remove the oil pressure relief valve (5), the gasket (8), and the rectangular seal ring (1).

Discard the gasket and seal.

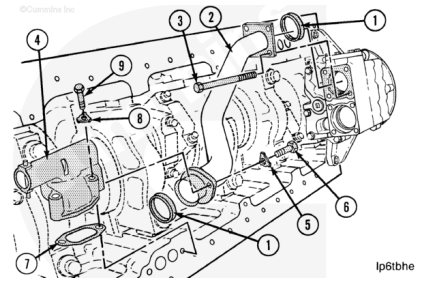
Remove the three capscrews (4), the tube (2), and the seal (1).

Discard the seal.



⚠ CAUTION ⚠

The capscrews (3) are special fatigue rated parts. Do not damage the parts by throwing them against each other or in a container. Damage to the parts can result in failure.



The Remove instructions for the K50 with High Speed (Three Gear) Pump are listed below:

Bend all of the lockplates from the heads of the capscrews.
Remove the two capscrews (6) and the two capscrews (9).
Remove the oil pressure relief valve (4), the gasket (7), and the rectangular seal ring (1).

Discard the gasket and seal.

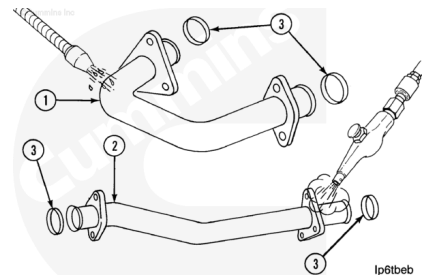
Remove the four capscrews (3), the tube (2), and the seal (1).

Discard the seal.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : The oil transfer tubes for the K38 and K50 engines are different. The procedure is the same. The K38 parts are illustrated as an example.

The tube (1) is for the oil filter head when it is located on the right bank of the engine. The tube (2) is for the oil filter head when it is located on the left bank of the engine.

Remove the rectangular seal rings (3) from the tube. Discard the rings.

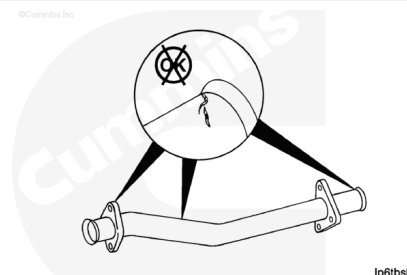
Use solvent or steam to clean the tube.

Dry with compressed air.

⚠ CAUTION ⚠

If the tube is repaired by welding, be sure all the welding slag and any metal that can break are removed from the inside of the tube. If any debris enters the lubricating oil pump and system, the gears will be damaged and failure will result.

Check the part for cracks or other damage. If the tube is cracked or damaged, it **must** be repaired or replaced.



Install

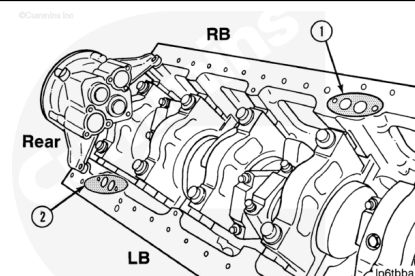
⚠ CAUTION ⚠

The correct oil transfer tubes must be installed or the engine will be damaged extensively from lack of lubrication.

If the oil filter head is to be installed on the left bank, "LB", the oil transfer tube **must** be connected to the main oil passages at the center of the right bank, "RB" (1).

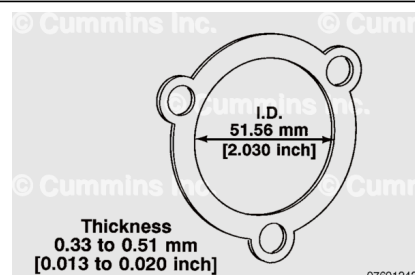
If the oil filter head is to be installed on the right bank, "RB", the oil transfer tube **must** be connected to the main oil passages at the rear of the left bank, "LB" (2).

The oil transfer tubes for the K38, the K50 with a low or high speed (two gear) pump, and the K50 with a high speed (three gear) pump are different. Following is the correct procedure for each.



⚠ CAUTION ⚠

If the gasket for the oil suction tube is installed between the oil pump and the oil pressure regulator housing, the gasket can blow out causing a leak that will reduce engine oil pressure and cause severe engine damage.



Note : The gasket (4) was previously red. New gaskets **must** be identified by gasket thickness and the inside diameter of the large hole in the center of the gasket.

Note : Do not tighten any of the capscrews until all of the parts are installed.

The Install instructions for the K38, Left Bank Lubrication Oil Filter Head are listed below:

Install the o-ring seals (1) on the transfer tube (2). Use vegetable oil to lubricate the seals.

Install the tube in the bore of the pressure relief valve (3).

Install the three lockplates (5) on the three capscrews (6).

Install the gasket (4), the oil pressure relief valve (3), the tube (2), and the lockplates and capscrews to the pump.

Install the oil pressure relief valve to the lubricating pump with the valve assembly angled toward the oil pan sump.

Install the connection (7) on the tube. Install the two lockplates (8) on the two capscrews (9). Install the two capscrews and lockplates through the tube flange to the connection.

Install the connection gasket (10). Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

Install the two lockplates (11) on the two capscrews (12). Install the lockplates and capscrews through the connection and gasket to the cylinder block.

Check to be sure the transfer tube (2) is **not** touching any of the parts of the cylinder block.

Push the connection (7) and the tube (2) toward the pump. Be sure the parts are aligned correctly.

Tighten the two capscrews (12).

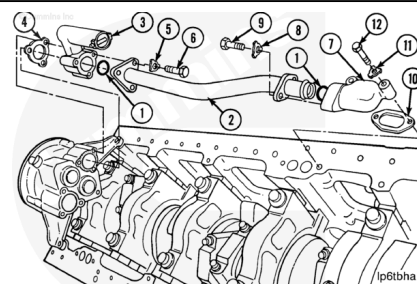
Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (9).

Torque Value: 40 n•m [30 ft-lb]

Tighten the three capscrews (6).

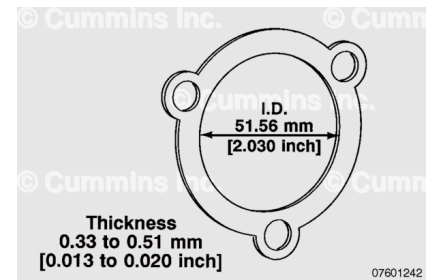
Torque Value: 40 n•m [30 ft-lb]



Check to be sure all parts are flat against the adjacent parts.
Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.

⚠ CAUTION ⚠

If the gasket for the oil suction tube is installed between the oil pump and the oil pressure regulator housing, the gasket can blow out causing a leak that will reduce engine oil pressure and cause severe engine damage.



Note : The gasket (4) was previously red. New gaskets **must** be identified by gasket thickness and the inside diameter of the large hole in the center of the gasket.

The Install instructions for the K38, Right Bank Lubricating Oil Filter Head are listed below:

Install the o-ring seals (1) on the transfer tube (2). Use vegetable oil to lubricate the seals.

Install the tube in the bore of the pressure relief valve (3).

Note : Do not tighten any of the capscrews until all of the capscrews are installed.

Install the three lockplates (5) on the three capscrews (6).

Install the gasket (4), the oil pressure relief valve (3), the tube (2), and the lockplates and capscrews to the pump.

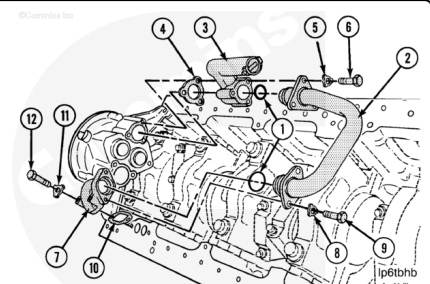
Note : Install the oil pressure relief valve to the lube pump with the valve assembly angled towards the oil pan sump.

Install the connection (7) on the tube. Install the two lockplates (8) on the two capscrews (9). Install the two capscrews and lockplates through the tube flange to the connection.

Install the connection gasket (10). Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

Install the two lockplates (11) on the two capscrews (12).

Install the lockplates and capscrews through the connection and gasket to the cylinder block.



Push the tube (2) toward the pump. While pushing the tube to the pump, push the connection (7) toward the tube and the cylinder block to be sure the parts are aligned correctly.

Tighten the three capscrews (6).

Torque Value: 40 n•m [30 ft-lb]

Tighten the two capscrews (12).

Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (9).

Torque Value: 40 n•m [30 ft-lb]

Check to be sure all the parts are flat against the adjacent parts. Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.

The K50 Low Speed Lubricating Oil Pump, Left Bank Lubricating Oil Filter Head installation instructions follow:

Install the rectangular ring seals (1) on the transfer tube (2).

Use vegetable oil to lubricate the seals.

Install the tube (2) into the smaller bore on the pump.

Note : Do not tighten any of the capscrews until all of the parts are installed.

Install the lockplates (3) on the capscrews (4). Install the capscrews through the flange on the tube to the pump.

Install the pressure relief valve (5) on the tube. Install the lockplates (6) on the capscrews (7). Install the flange on the tube to the valve.

Install the gasket (8) between the valve and the cylinder block. Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

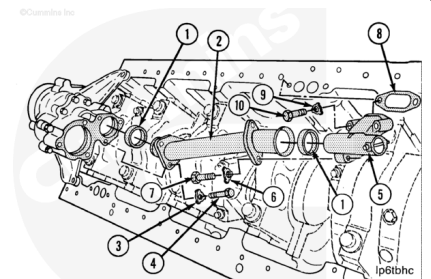
Install lockplates (9) on the two capscrews (10). Install the capscrews (10) through the valve and gasket to the cylinder block.

Check to be sure that the transfer tube (2) is **not** touching any of the parts of the cylinder block. Be sure the flanges on the tube are flat against the adjacent parts.

Tighten the three capscrews (4).

Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (10).



Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (7).

Torque Value: 40 n•m [30 ft-lb]

Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.

The K50 Low Speed Lubricating Oil Pump, Right Bank Lubricating Oil Filter Head installation instructions follow:

Install the rectangular ring seals (1) on the transfer tube (2).

Use vegetable oil to lubricate the seals.

Install the tube (2) into the smaller bore on the pump.

Note : Do not tighten any of the capscrews until all of the parts are installed.

Install the lockplates (3) on the capscrews (4). Install the capscrews through the flange on the tube to the pump.

Install the pressure relief valve (5) on the tube. Install the lockplates (6) on the capscrews (7). Install the flange on the tube to the valve.

Install the gasket (8) between the valve and the cylinder block. Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

Install the lockplates (9) on the two capscrews (10). Install the capscrews (10) through the valve and gasket to the cylinder block.

Be sure the flanges on the tubes are flat against the adjacent parts.

Push the tube (2) toward the pump. While pushing the tube to the pump, push the valve (4) toward the tube and the cylinder block to be sure the parts are aligned correctly.

Tighten the three capscrews (4).

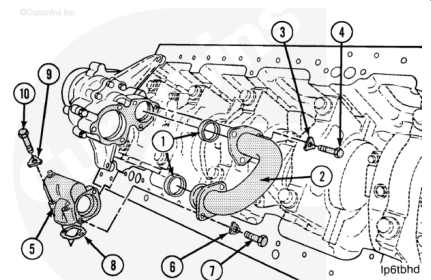
Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (10).

Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (7).

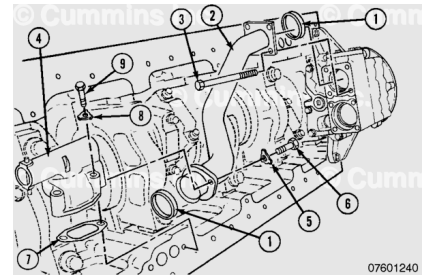
Torque Value: 40 n•m [30 ft-lb]



Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.

⚠ CAUTION ⚠

The capscrews (3) are special fatigue-rated capscrews, Part Number 3026676. The part number is stamped on the head of the capscrew. Do not install any other capscrews or the part will fail. Use caution when handling the capscrews to prevent damage.



The K50 High Speed Lubricating Oil Pump, Left Bank
Lubricating Oil Filter Head installation instructions follow:

Install the rectangular ring seals (1) on the transfer tube (2).

Use vegetable oil to lubricate the seals.

Install the tube (2) into the bore on the pump nearest to the left bank.

Note : Do not tighten any of the capscrews until all of the parts are installed.

Install the four capscrews (3) through the tube flange to the pump.

Install the pressure relief valve (4) on the tube. Install the lockplates (5) on the capscrews (6). Install the capscrews through the flange on the tube to the valve.

Install the gasket (7) between the valve and the cylinder block. Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

Install the lockplates (8) on the capscrews (9).

Install the capscrews (9) through the valve and gasket to the cylinder block.

Check to be sure the tube (2) is **not** touching any of the parts of the cylinder block. If there is contact, turn the tube.

Be sure that the flanges on the tube are flat against the adjacent parts.

Push the valve (4) and the tube (2) toward the pump. Be sure the flanges on the tubes are flat against the adjacent parts.

Tighten the four capscrews (3).

Torque Value: 95 n•m [70 ft-lb]

Tighten the two capscrews (9).

Torque Value: 60 n•m [44 ft-lb]

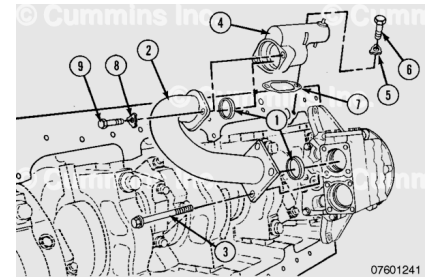
Tighten the two capscrews (6).

Torque Value: 40 n•m [30 ft-lb]

Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.

⚠ CAUTION ⚠

The capscrews (3) are special fatigue-rated capscrews, Part Number 3026676. The part number is stamped on the head of the capscrew. Do not install any other capscrews or the part will fail. Use caution when handling the capscrews to prevent damage.



The Install instructions for the K50 High Speed Lubricating Oil Pump, Right Bank, Lubricating Oil Filter Head are listed below:

Install the rectangular ring seals (1) on the transfer tube (2).

Use vegetable oil to lubricate the seals.

Install the tube (2) into the bore on the pump nearest to the left bank.

Note : Do not tighten any of the capscrews until all of the parts are installed.

Install the four capscrews (3) through the tube flange to the pump.

Install the pressure relief valve (4) on the tube. Install the lockplates (5) on the capscrews (6). Install the capscrews through the flange on the tube to the valve.

Install the gasket (7) between the valve and the cylinder block. Install the outward embossed face of the gasket towards the transfer block face and the flat back face of the gasket towards the engine block.

Install the lockplates (8) on the capscrews (9). Install the capscrews (9) through the valve and gasket to the cylinder block.

Be sure the flanges on the tube are flat against the adjacent parts.

Push the tube (2) toward the pump. While pushing the tube to the pump, push valve (4) toward the tube and the cylinder block to be sure the parts are aligned correctly.

Tighten the four capscrews (3).

Torque Value: 95 n•m [70 ft-lb]

Tighten the two capscrews (9).

Torque Value: 60 n•m [44 ft-lb]

Tighten the two capscrews (6).

Torque Value: 40 n•m [30 ft-lb]

Bend the lockplates over the parts and the capscrew heads to be sure the capscrews do **not** loosen.